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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

MA2007-1 – BIOSTATISTICS
(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – II & Unit – III each** and **One full** question from **Unit – I**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The median for the data: 1, 0, -2, 7, 4, 8, -3 is
A) 7
B) 0
C) 9
D) 1
2. The range for the data: 61, 12, 0, -5, -1,, 60, 59 is
A) 66
B) 60
C) 61
D) 59
3. The G.M. of 64, 9, 31, 18 is
A) 30.5
B) 23.81
C) 80352
D) 23
4. If the mean and mode of a distribution are 38 and 32 respectively, then median is
A) 32
B) 35
C) 38
D) 36
5. $E\left(\frac{1}{6}\right)$ is
A) $\frac{1}{6}$
B) 0
C) $\frac{1}{36}$
D) $-\frac{1}{6}$
6. The value of cumulative distribution function $F(+\infty)$ is
A) -1
B) 0
C) $+\infty$
D) 1
7. For a discrete random variable $\sum P(x)$ is equal to
A) 0
B) -1
C) 1
D) Not defined
8. If $V(X)=2$ then $V(2X+3)$ is
A) 4
B) 8
C) -11
D) 7
9. A standard normal curve has the mean and variance:
A) (1,1)
B) (1,0)
C) (μ, σ^2)
D) (0,1)
10. The variance of Poisson distribution with parameter 2 is
A) 1
B) 2
C) 0
D) -2
11. The mean of a binomial distribution is
A) npq
B) $\sqrt{npq}$
C) $\sqrt{np}$
D) np

12. The only distribution in which the mean, median and mode coincide is
 A) normal B) geometric
 C) binomial D) poisson
13. For a 3x4 contingency table, the number of d.o.f. is
 A) 6 B) 12
 C) 8 D) 0
14. A measure concerned with a population is called
 A) Estimate B) parameter
 C) estimation D) statistic
15. To represent null hypothesis, we use the notation
 A) H_0 B) H_1
 C) H_2 D) h_2
16. The probability of rejecting a hypothesis when it is false is
 A) Type I error B) Type II error
 C) Type III error D) power of a test
17. Sampling is the process of
 A) learning about the sample B) learning about the population
 C) either of these D) none of these
18. The more suitable sampling procedure for a heterogeneous population is
 A) Simple random sampling B) stratified sampling
 C) multistage sampling D) purposive sampling
19. Analysis of variance technique was developed by
 A) W.S. Gosset B) R. A. Fisher
 C) De-Moivre D) Karl-Pearson
20. The number of d.o.f. for total sum of squares is:
 A) $r-1$ B) n
 C) $n-r$ D) $n-1$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Construct a histogram and a frequency polygon for the following data.

class	100-150	150-200	200-250	250-300	300-350
f	4	6	13	5	2

Marks BT* CO* PO*

5 L*1 1 2

- b) Write the formula for G.M. for a raw data. The data on monthly income of 70 persons are given below. Calculate the harmonic mean.

Income	0-10	10-20	20-30	30-40	40-50
Persons	8	14	19	17	12

5 L2 1 1

- c) The profits earned by 100 companies during 1989-90 are given below. Compute the standard deviation of the profits.

Profits	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of companies	4	8	18	30	15	10	8	7

6 L3 1 1

2. a) Determine the coefficient of variation for the data:
 44, 80, 76, 48, 52, 72, 68, 56, 60, 54.

5 L1 1 1

- b) Calculate the median and mode of the following data.

Weight	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of babies	15	20	25	24	10	33	71	51

5 L3 1 2

- c) Draw the ogive's for the following data and locate the median.

Marks	0-4	4-8	8-12	12-16	16-20
No. of students	4	6	10	8	4

6 L2 1 1

Unit – II

3. a) In a certain college 4% of boys and 1% of girls are taller than 1.8m. Also 60% of the students are girls. If a student is selected at random and found taller than 1.8m, what is the probability that the student is a girl?
 b) Obtain the mean and variance of a Poisson distribution.
 c) The pdf of $P(X)$ of a variate X is given by the following table

X	0	1	2	3	4	5	6
P(X)	k	3k	5k	7k	9k	11k	13k

For what value of k , does this represent a valid probability distribution? Also find $P(x \leq 2)$, $P(1 \leq x \leq 5)$ and $E(3X + 1)$

5 L2 2 1
5 L2 3 2

6 L1 2 2

4. a) Given $f(x) = kx^3$, $0 < x < 1$
 0 else where

Find (i) value of k such that $f(x)$ is a valid pdf.

ii) find $E(X)$

5 L1 2 1

- b) A fair coin is tossed 3 times. Let X denote the number of heads obtained. Find the distribution of X . Also find its mean and variance.
 c) In a test on 2000 electrical bulbs, it was found that the life of a particular bulb are normally distributed with average life of 2040 hours and S.D. of 60hrs. Estimate the number of bulbs likely to burn for i) more than 2150hrs ii) less than 2000hrs.
5. a) Prove that $V(X) = E(X^2) - [E(X)]^2$ Also prove that $E(aX + b) = aE(X) + b$ and $V(aX + b) = a^2V(X)$
 b) The daily turn over in a medical shop is exponentially distributed with Rs. 6000 as the average. Find the probability that the turn over will exceed Rs. 6250 on a randomly chosen day.
 c) If the number of accidents occurring on a highway each day is a Poisson variate with mean equal to 4. What is the probability that
 i) 3 or more accidents occur ii) no accidents occur on a random day.

5 L2 2 2

6 L2 3 2

5 L3 2 2

5 L2 3 1

6 L2 3 1

Unit – III

6. a) In a city A 20% of random sample of 900 school boys had a certain psychological defect. In another city B 18.5% of random sample of 1600 school boys had the same defect. Is the difference between the proportions significant based on 5% level of significance.
 b) The following table show the number of students in each of 2 classes A and B who passed and failed in an examination. Test the hypothesis using Chi square test that there is no difference in classes in respect of passing or failing in the examination.

	passed	failed
Class A	72	17
Class B	64	23

7 L3 4 1

9 L2 4 2

7. a) Find 95% confidence interval for the mean from the data: -4, -2, -2, 0, 2, 2, 3, 3. (Given t value at 95% confidence level for 7 df=3.5).
- b) Use F-test and analyze if there is any significant difference in the yield per acre at 5% level for the data.

A	8	4	6	7
B	7	5	5	3
C	2	5	4	4

8. a) Write a short note on simple random sampling and stratified random sampling.
- b) One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significant difference in the two types of aircrafts with regard to engine defects? (Use 5% level).
- c) Among the offspring of a certain cross between guinea pigs 34 were red, 10 were black and 20 were white. According to the genetic model these numbers should be in the ratio 9:3:4. Are the data consistent with the model at 5% level?

6	L2	4	2
10	L3	5	1
4	L2	5	2
6	L3	4	1
6	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
